# CS & IT

## Theory of computation

DPP: 1

## CFL

**Q1** Consider the following grammar:
S → abSba | bSb | aSa | a | b | ∈
What is the language generated by above grammar?
(A) CFG that generates all palindromes over alphabet (a, b).
(B) CFG that generates all palindromes over alphabet (a, b) that do not contain substring "ab".
(C) CFG that generates all palindromes over alphabet (a, b) that contain substring "aa".
(D) CFG that generates all palindromes over alphabet (a, b) that do not contain substring "aa".

**Q2** Consider the following grammars:
**$G_1$:** S → aSc | $ B
B → bBc | $
**$G_2$:** S → aaScc | $ B
B → bBc | $
**$G_3$:** S → aSc | $ B
B → bbBcc | $
Which of the following is true regarding $G_1$, $G_2$ and $G_3$?
(A) $L(G_1) \subseteq L(G_2)$, $\subseteq(G_2) \subseteq L(G_3)$
(B) $L(G_2) \subseteq L(G_3)$, $L(G_1) \subseteq L(G_3)$
(C) $L(G_1) \subseteq L(G_3)$, $L(G_3) \subseteq L(G_2)$
(D) $L(G_2) \subseteq L(G_1)$, $L(G_3) \subseteq L(G_1)$

**Q3** Consider the following context free grammar:
S → aA | aBB
A → aaA | ∈
B → bB | bbC
C → B
What will be the equivalent simplified CFG for the given grammar?
(A) S → aA | a
A → aaA | aa | b
(B) S → aA | a
A → aaA | aa
(C) S → aAa | B
A → aaA | aa
B → bB | bb
(D) S → aAa | B
A → aA | b
B → bB | bb | a

**Q4** Consider the following grammar:
S → AB
A → BaB | a
B → bbA
Which of the following is true regarding given grammar?
(A) Every string of the above grammar have at least two a's.
(B) Every string have three consecutive a's.
(C) Every string have alternate a and b.
(D) Every string have b's in multiple of 2.

**Q5** Given the following two grammars:
**$G_1$:** S → AB | aaB
A → a | Aa
B→ b
**$G_2$:** S → aSbS | bSaS | ∈
What is true regarding above grammars?
(A) Both $G_1$ and $G_2$ are ambiguous.

(B) Both $G_1$ and $G_2$ are unambiguous.
(C) Only $G_1$ is ambiguous.
(D) Only $G_1$ is unambiguous.

**Q6** Consider the following languages
$L_1 = \{a^n b^{n+m} c^m \mid n, m \geq 0\}$
$L_2 = \{a^p b^q c^{p+q} \mid p, q \geq 0\}$
Which of the following is true?
(A) Only $L_1$ is Regular.
(B) Only $L_2$ is Regular.
(C) Both $L_1$ and $L_2$ are Regular.
(D) None of these

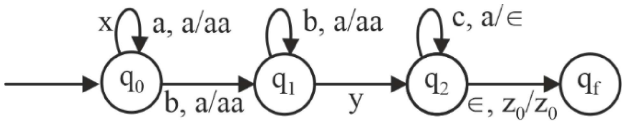
**Q7** Which of the following is string accepting mechanism of PDA?
(A) PDA using final state.
(B) PDA using empty stack.
(C) PDA using both empty stack and final state.
(D) PDA using transition state.

**Q8** Which of the following is correct push operation:
(A) $\Sigma$ (q, a , b) = (q', ab)
(B) $\delta$ (q, a , b) = (q, ab)
(C) $\delta$ (q, a , b) = (q', ab)
(D) $\Sigma$ (q, a , b) = (q',$\in$)

**Q9** Consider the following statements:
(I) All DPDA are NPDA.
(II) All NPDA are DPDA.
(III) All NPDA and DPDA are equivalent.
(IV) All context free language are regular language.
The total number of correct statements are_____.

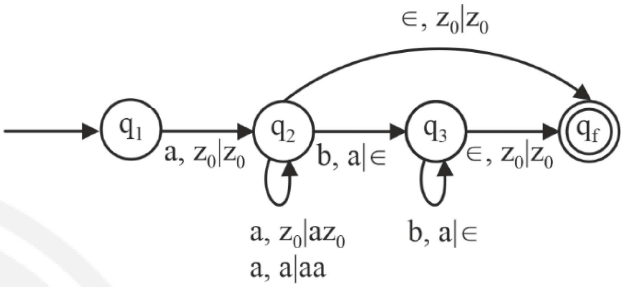
**Q10** What does following transition means:
$\delta(q, \in, b) = (q', b)$
(A) Push b (B) Pop b
(C) Read b (D) No operation

**Q11** What are the values of x and y, if the language accepted by NPDA is $L = \{a^m b^n c^{(m+n)} \mid m, n \geq 1\}$.




(A) x = $\in$, a/a ; y = c, b/$\in$
(B) x = a, $z_0/z_0$ ; y = c, b/c
(C) x = a, a/aa ; y = b, c/c
(D) x = a, $z_0/az_0$ ; y = c, a/$\in$

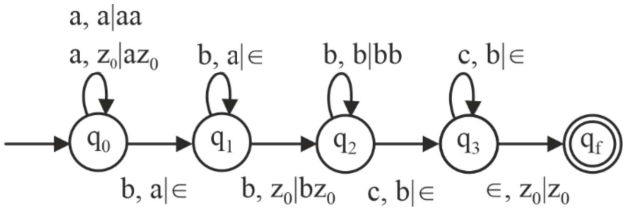
**Q12** Which language is accepted by the following PDA.




(A) $L = \{a^n b^{n+1} \mid n \geq 1\}$
(B) $L = \{a^{n+1} b^m \mid n, m \geq 1\}$
(C) $L = \{a^{n+1} b^n \mid n \geq 0\}$
(D) $L = \{a^n b^{n+1} \mid n \geq 0\}$

**Q13** Which of the following languages are accepted by PDA.
(A) $L = \{a^n b^n c^m \mid m, n \geq 1\}$
(B) $L = \{a^n b^n c^m \mid m \leq n\}$
(C) $L = \{a^n b^m c^n d^m \mid m, n \geq 0\}$
(D) $L = \{a^m b^n c^n d^m \mid m, n \geq 0\}$

**Q14** The language derived by given PDA is




(A) $L = \{a^n b^m c^k \mid n = m + k, m, n, k \geq 0\}$.
(B) $L = \{a^m b^n c^k \mid k = m + n, m, n, k \geq 1\}$.
(C) $L = \{a^m b^n c^k \mid k = m + n, m, n, k \geq 0\}$.
(D) $L = \{a^m b^n c^k \mid n = m + k, m, n, k \geq 1\}$.

**Q15**

Suppose $L_1$ is a finite language and $L_2$ is non-regular language then $L_1 \cap L_2$ will be:
(A) Regular but infinite.
(B) Non-regular.
(C) Finite and regular.
(D) None of these.

**Q16** Consider following statements:
$S_1$: Kleene Closure of infinite set is always finite.
$S_2$: Kleene Closure of finite set is always infinite.
Which of the following is correct?
(A) $S_1$ only (B) $S_2$ only
(C) Both $S_1$ and $S_2$ (D) None of these

**Q17** Consider a language L (a+b)*, then subset of L will be?
(A) Regular
(B) Regular but finite
(C) Non-regular
(D) None of these

**Q18** Let $L_1$ be a regular language and $L_2$ be a CFL language, then which of the following is correct?
(A) $L_1 \cap L_2$ is CFL
(B) $L_1 - L_2$ must be regular
(C) $L_1 \subset L_2$ must be regular
(D) None of these.

**Q19** Consider two languages $L_1$ and $L_2$.
$L_1 \subset L_2$, $L_2$ is regular then which of the following is true?
(A) $L_1$ must be finite
(B) $L_1$ must be regular
(C) $L_1$ must be non-regular
(D) None of these

**Q20** Regular language does not closed under on which operation?
(A) Complement (B) Union
(C) Subset (D) Intersection.

# Answer Key

| | |
|---|---|
| Q1 | (A) |
| Q2 | (D) |
| Q3 | (B) |
| Q4 | (A, D) |
| Q5 | (A) |
| Q6 | (D) |
| Q7 | (A, B, C) |
| Q8 | (B, C) |
| Q9 | 1 |
| Q10 | (D) |
| Q11 | (D) |
| Q12 | (C) |
| Q13 | (A, D) |
| Q14 | (D) |
| Q15 | (C) |
| Q16 | (D) |
| Q17 | (D) |
| Q18 | (A) |
| Q19 | (D) |
| Q20 | (C) |

# Hints & Solutions

**Q1 Text Solution:**

Given

$S \rightarrow abSba \mid bSb \mid aba \mid a \mid b \mid \hat{I}$

Strings formed using above grammar are:

{∈, a, b, aa, bb, .....}

This is the CFG that generates all palindromes over alphabet (a, b).

So, option (a) is correct answer.

**Q2 Text Solution:**

$L(G_1) = \{a^m \$ b^n \$ c^{m+n} \mid m, n \geq 0\}$

$L(G_2) = \{a^{2m} \$ b^n \$ c^{2m+n} \mid m, n \geq 0\}$

$L(G_3) = \{a^m \$ b^{2n} \$ c^{m+2n} \mid m, n \geq 0\}$

∴ It is clear,

$L(G_2) \subseteq L(G_1)$, and

$L(G_3) \subseteq L(G_1)$

Therefore, option d is correct.

**Q3 Text Solution:**

Simplified CFG is a CFG without any null productions, unit-productions and useless symbol.

1. Elimination of NULL Productions:

   $A \rightarrow \in$

   Keep all the productions as it is and substitute ∈ in place of A.

   $S \rightarrow aA \mid aBB \mid a$

   $A \rightarrow aaA \mid aa$

   $B \rightarrow bB \mid bbC$

   $C \rightarrow B$

2. Eliminate Unit Productions: If the productions is of the form $A \rightarrow B$, the production is known as unit production. Eliminate unit production by replacing equivalent derivations.

   $S \rightarrow aA \mid aBB \mid a$

   $A \rightarrow aaA \mid aa$

   $B \rightarrow bB \mid bbB$

3. Eliminate useless symbol: The variables that are not involved the derivation of any string is known as useless symbol.

   In this question B is deriving bB | bbB which is entering into infinite loop. So, eliminating all productions of B.

   $S \rightarrow aA \mid a$

   $A \rightarrow aaA \mid aa$

   Hence, options (b) is the correct answer.

**Q4 Text Solution:**

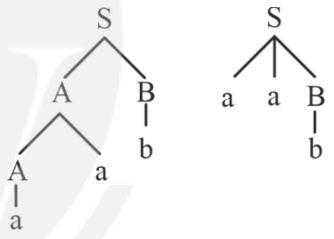
The string produced by given grammar is {abba, abbaabba, ……}

So option (a) and (d) are the true statements.

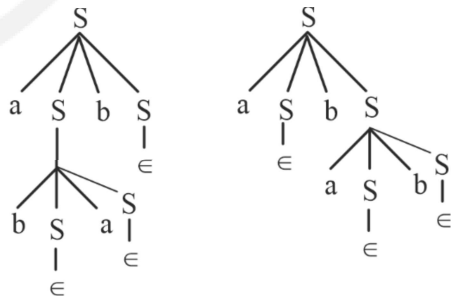
**Q5 Text Solution:**

Both $G_1$ and $G_2$ are ambiguous, as both have multiple parse tree for same string.

**$G_1$:** aab




**$G_2$:** abab




**Q6 Text Solution:**

$L_1 = \{\in, ab, bc, aabb, bbcc, \ldots\ldots\}$

$L_2 = \{\in, ac, bc, aacc, bbcc, \ldots\ldots\}$

Both the languages are CFL not regular.

Therefore option (d) is correct.

**Q7 Text Solution:**

PDA accepts string using final state, empty stack & both and all methods are equivalent.

**Q8 Text Solution:**

$\delta(q, a, b) = (q, ab)$ is correct push operation on current state.

$\delta(q, a, b) = (q', ab)$ is the correct push operation

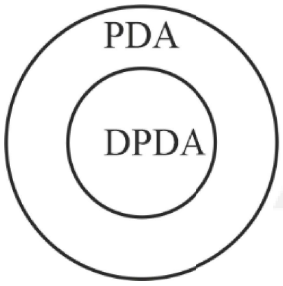
$\delta(q, a, b) = (q', ab)$



Push a

**Q9 Text Solution:**

(I) All DPDA are PDA: correct




(II) All PDA are DPDA: incorrect

Some PDA are DPDA but all PDA are not DPDA.

(III) All PDA and DPDA care equivalent: incorrect

DPDA are subset of PDA. So all PDA and DPDA are not equivalent.

(IV) All context free language are regular language: incorrect

CFL are superset of RL. So, this statements is incorrect. However, all regular languages are context free languages.

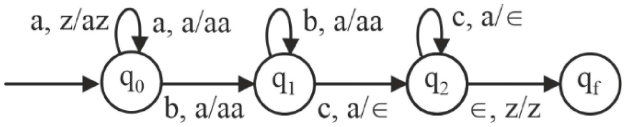
Therefore only 1 statement is correct.

**Q10 Text Solution:**

$\delta(q, \in, b) = (q', b)$, this represents no operation is performed on stack. So, there is no change in the stack.

**Q11 Text Solution:**

The PDA will be:




The x is to insert/push initial "a" into stack.

The y is to remove/pop an "a" for each "c".

**Q12 Text Solution:**

$L = \{a^{n+1} b^n \mid n \geq 0\}$

$= \{a^1 \ a^n \ b^n \mid n \geq 0\}$

↓ ↓ ↓

Skip Push Pop

In the given PDA, first "a" is being skip. Then, for "n' number of 'a' are inserted and once 'b' starts appearing for each 'b', we will pop one "a".

Hence, option (c) is correct.

**Q13 Text Solution:**

(a) $L = \{a^n b^n c^m \mid m, n \geq 1\}$

Push 'n' no of 'a'. pop 'a' for each 'b', skip 'c'.

It is accepted by PDA.

(b) $L = \{a^n b^n c^m \mid m \leq n\}$

Push 'n' no of 'a', pop 'a' for each 'b'. Since, it is given that m £ n, so we can't compare number of c. Therefore, it is not acceptable using PDA.

(c) $L = \{a^n b^m c^c d^m \mid m, n \geq 0\}$

Unable to compare.

Therefore, it is not accepted by PDA.

(d) $L = \{a^m b^n c^n d^m \mid m, n \geq 0\}$

Push 'm' no. of "a". Push 'n' no. of "b".

Pop 'b' for each 'c'. Pop 'a' for each 'd'.

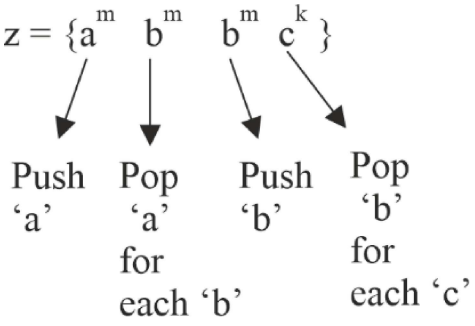
Hence, it is accepted by PDA.

**Q14 Text Solution:**

The given PDA will generate language

$L = \{a^m b^n c^k \mid n = m + k, m, n, k \geq 1\}$

$L = \{a^m b^{m+k} c^k\}$




**Q15** **Text Solution:**

Finite ∩ non-regular always finite.

Hence, option (c) is correct.

**Q16** **Text Solution:**

**$S_1$:** False

Set = $\{\in\} = \{\in\}^* = \in$ only (Finite)

**$S_2$:** Set = $\{a\} = \{a\}^* = \in, a, aa, aaa, .. = (a^*)$ (Infinite)

So, both statements are false.

Hence, option (d) is correct.

**Q17** **Text Solution:**

$(L) = (a + b)^*$

- $a^n b^n$ is a subset of $(a + b)^*$
  but $a^n b^n$ is not a regular and also not finite.
- ab is a subset of L but ab is a finite and regular.

Hence, option (d) is correct.

**Q18** **Text Solution:**

$L_1$ = Regular

$L_2$ = CFL

(a) Regular ∩ CFL Need not be regular.

Example:

$= (a + b)^* (L_1) \cap a^n b^n (L_2)$

$= a^n b^n$(CFL).

(b) Regular – CFL

= Regular ∩ $\overline{\text{CFL}}$

= Need not be regular

(c) Regular ⊂ CFL

Need not be regular

Hence, option (d) is correct.

**Q19** **Text Solution:**

If $L_1 \subset L_2$, then $L_1$ can be finite or infinite

Hence, option (d) is correct.

**Q20** **Text Solution:**

Subset of regular language need not be regular